The *Bisection* Method

Introduction

- Bisection Method:

Bisection Method = a numerical method in Mathematics to find a root of a given *function*

Introduction (cont.)

Root of a function:

Root of a function $f(x)$ = a value a such that:

$$f(a) = 0$$

Introduction (cont.)

Example:

Function: $f(x) = x^2 - 4$

Roots: $x = -2,\ x = 2$

Because:

$$f(-2) = (-2)^2 - 4 = 4 - 4 = 0$$
$$f(2) = (2)^2 - 4 = 4 - 4 = 0$$

A Mathematical Property

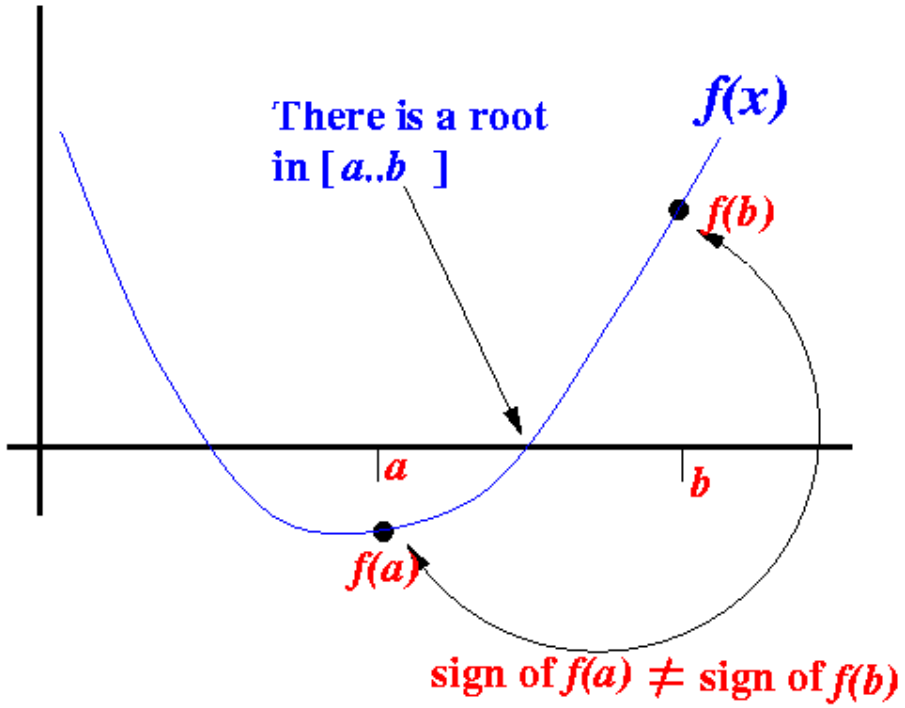
Well-known Mathematical Property:

If a function $f(x)$ is continuous on the interval $[a..b]$ and sign of $f(a) \neq$ sign of $f(b)$, then

There is a value $c \in [a..b]$ such that: $f(c) = 0$ i.e., there is a root c in the interval $[a..b]$

A Mathematical Property (cont.)

Example:



The *Bisection* Method

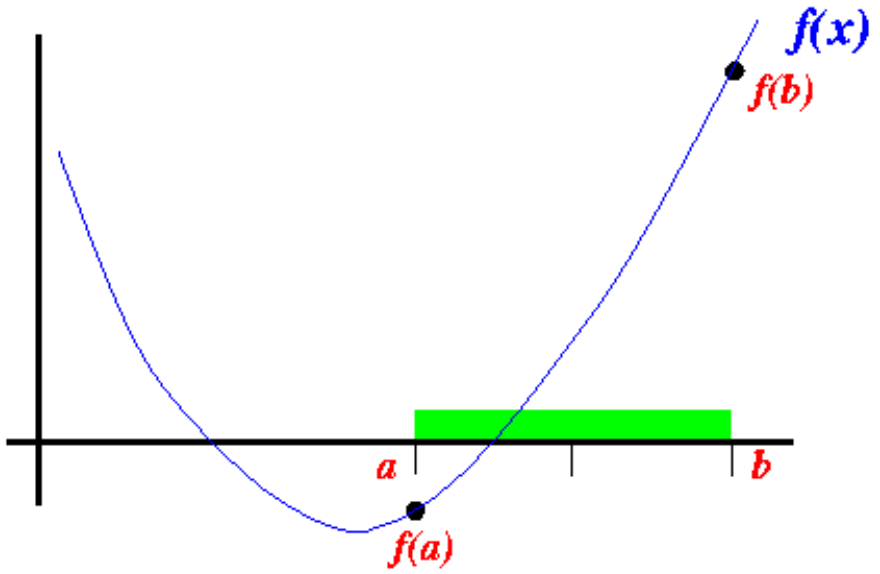
- The Bisection Method is a *successive approximation method* that narrows down an interval that contains a root of the function *f(x)*
- The Bisection Method is *given* an initial interval [*a*..*b*] that contains a root (We can use the property sign of *f(a)* ≠ sign of *f(b)* to find such an initial interval)
- The Bisection Method will *cut the interval* into 2 halves and check which half interval contains a root of the function
- The Bisection Method will keep *cut the interval* in halves until the resulting interval is extremely small

  The root is then *approximately equal* to *any value* in the final (very small) interval.

The *Bisection* Method (cont.)

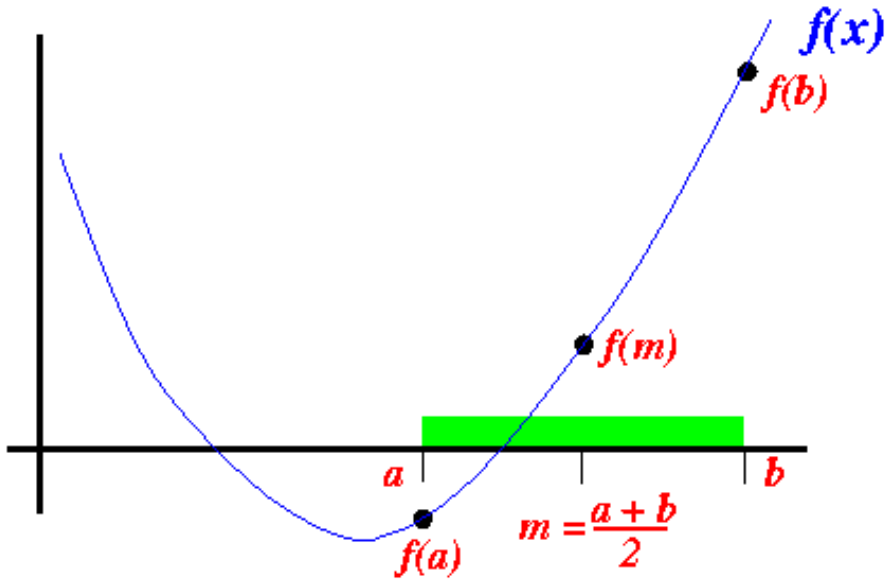
- Example:

Suppose the interval $[a..b]$ is as follows:



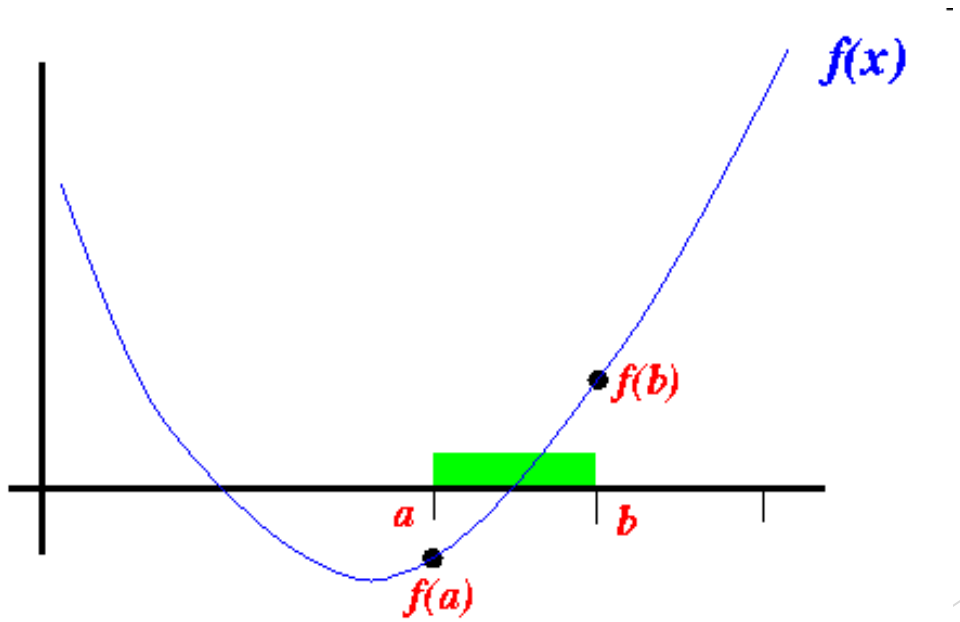
The *Bisection* Method (cont.)

We cut the interval [*a*..*b*] in the middle: *m* = *(a+b)/2*



The *Bisection* Method (cont.)

Because sign of $f(b)$ ≠ sign of $f(a)$, we *proceed* with the search in the *new* interval [*a*..*b*]:



The *Bisection* Method (cont.)

We can use this statement to change to the new interval:

```
b = m;
```

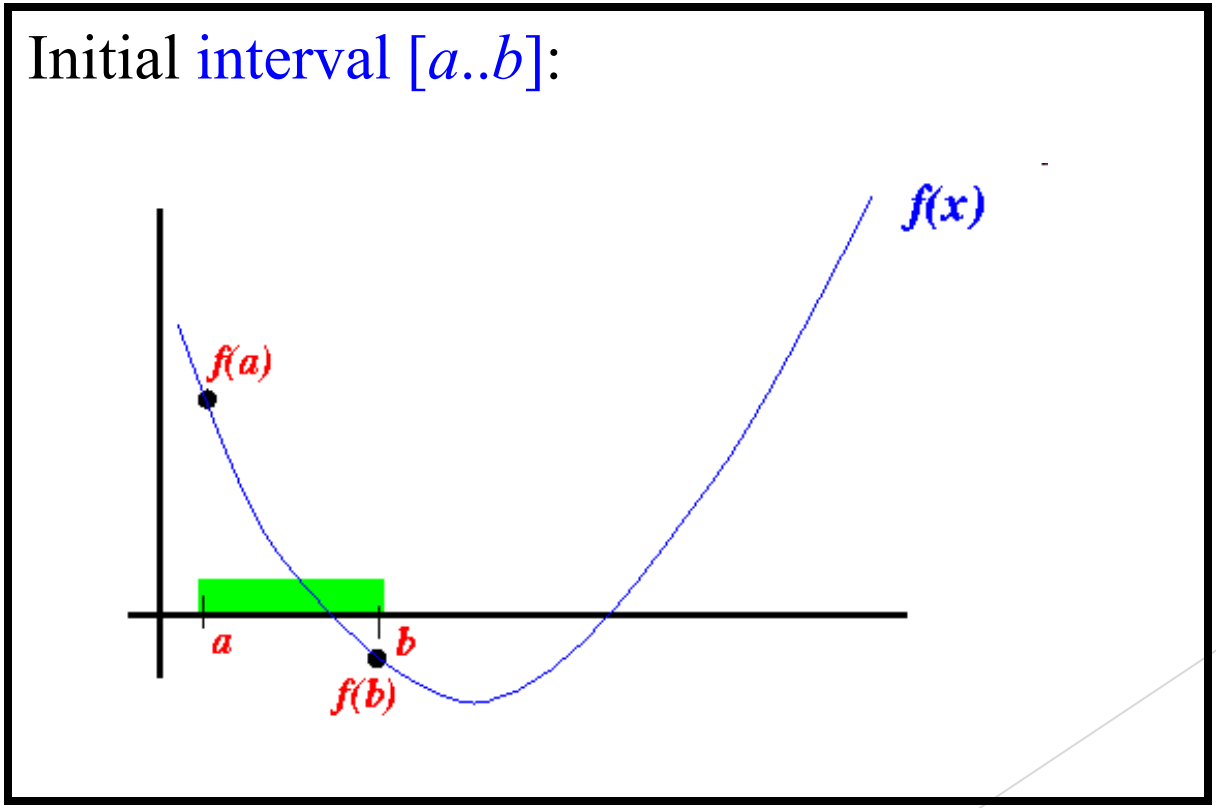
The *Bisection* Method (cont.)

In the above example, we have changed the end point *b* to obtain a smaller interval that still contains a root

In other cases, we may need to changed the end point *a* to obtain a smaller interval that still contains a root

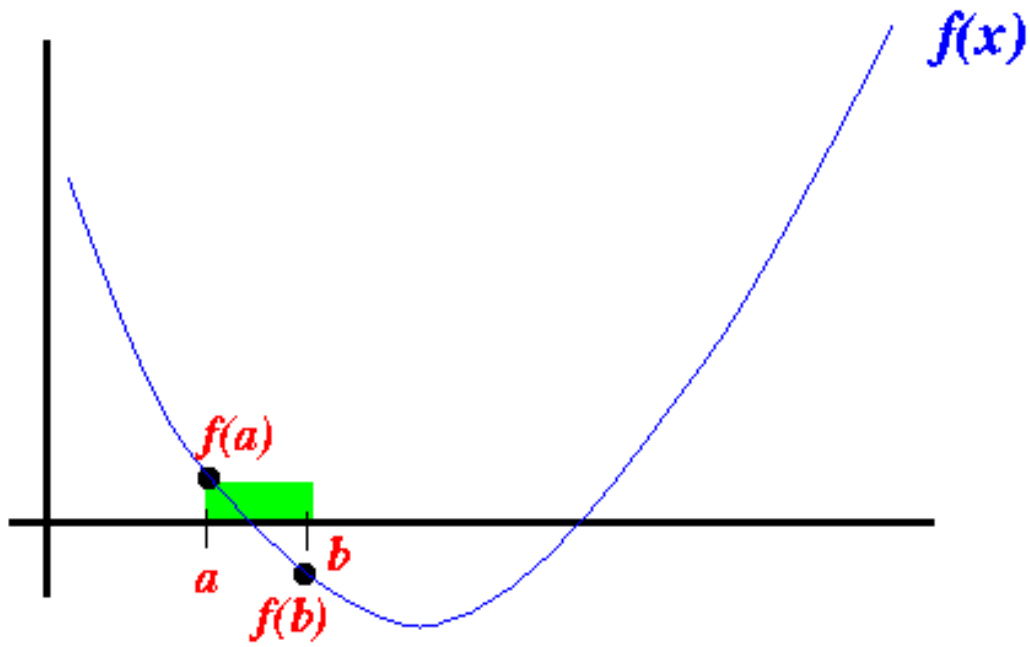
The *Bisection* Method (cont.)

- Here is an example where you have to change the end point a:



The *Bisection* Method (cont.)

After cutting the interval in half, the root is contained in the right-half, so we have to change the end point *a*:



The *Bisection* Method (cont.)

Rough description (pseudo code) of the Bisection Method:

1. Start with an interval $[a, b]$ such that $f(a) \cdot f(b) < 0$.
2. Calculate the midpoint $c = \frac{a+b}{2}$.
3. Evaluate $f(c)$:
   - If $f(c) = 0$, then c is the root.
   - Else if $f(a) \cdot f(c) < 0$, the root is in $[a, c]$, so set $b = c$.
   - Else the root is in $[c, b]$, so set $a = c$.
4. Repeat steps 2-3 until the interval width $|b - a|$ is less than a chosen tolerance.

The *Bisection* Method (cont.)

Assignment

Write a bisection method code in Python

Finding the roots of a *different* function

- Suppose we want to find the root of

$$f(x) = x^3 + x - 3$$

in the interval [0..4]

Quiz

In Next Class

Summary

- Reliable and simple method for root finding
- Requires sign change & continuity
- Converges linearly with predictable error bounds
- Use as initial step in root-finding